ELLA BOURASSA

M.S. COMPUTER SCIENCE STUDENT • MACHINE LEARNING • HUMAN-CENTERED COMPUTING

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PROFILE

M.S. Computer Science student at NYU Tandon with a B.S. in Cognitive Science and a computational emphasis from UC Davis. Interdisciplinary background in machine learning, algorithms, data structures, statistical modeling, cognitive science, and human-centered technology. Experienced in Python, C, C++, MATLAB, PyTorch, data analysis, technical testing, and communicating technical work clearly.

EDUCATION

Master of Science in Computer Science | New York University, Tandon School of Engineering **Expected May 2028**
Brooklyn, NY

Graduate coursework in progress: Design and Analysis of Algorithms I, Programming Languages, Big Data

Bachelor of Science in Cognitive Science, Computational Emphasis | University of California, Davis **June 2026**
Davis, CA | Cumulative GPA: 3.659

Selected coursework: Artificial Intelligence, Data Structures, Algorithm Design and Analysis, Theory of Computation, Computer Organization, Discrete Mathematics, Probability and Statistical Modeling, Linear Algebra, MATLAB Laboratory, and Neuroeconomics and Decision Making

TECHNICAL SKILLS

Languages: Python, C, C++, x86 assembly, MATLAB
Libraries and environments: PyTorch, Gymnasium, pandas, scikit-learn, Jupyter
Development tools: Git/GitHub, VS Code, CMake, GoogleTest, Valgrind, gdb
Methods: Machine learning, reinforcement learning, statistical modeling, model evaluation, data visualization, data structures and algorithms

SELECTED TECHNICAL PROJECTS

ChurnScope - Telecom Churn and Retention Modeling | Machine Learning Project **2026**
Python, pandas, scikit-learn

- Built and evaluated classification models on a telecom churn dataset; the best Random Forest model achieved approximately 0.679 ROC-AUC.
- Translated model probabilities into threshold and contact-capacity profit analyses, identifying when broad outreach was profitable and when constrained outreach was not.

SARSA Agent for Atari Pong | Reinforcement Learning Project **2025**
Python, PyTorch, Gymnasium ALE Pong-v5

- Implemented a SARSA agent using grayscale 84 x 84 observations, four-frame stacking, and an epsilon-greedy exploration schedule.
- Tracked training reward and compared results with DQN baselines, documenting limitations and potential improvements to representation, replay, and training duration.

PROFESSIONAL EXPERIENCE

Front Desk Staff / Shift Lead | Kidplex LLC **Jan-Mar 2025**
San Ramon, CA | Promoted to shift lead for February-March 2025

- Supervised staff, delegated daily work, coordinated schedules and parties, and maintained customer safety and smooth operations during events of up to 200 guests.
- Built a Google Apps Script inventory tool and analyzed Roller POS sales data to create a dynamic pricing model supporting operational decisions.

PROFESSIONAL EXPERIENCE, CONTINUED

Outdoor Gear Rental Associate | UC Davis Campus Recreation - Outdoor Adventures **Jun 2025-Jun 2026**
Davis, CA

- Guided customers through equipment selection, rental procedures, and safe gear use while delivering attentive service in a fast-paced environment.
- Processed payments, reservations, and orders through a point-of-sale system and maintained accurate rental records.
- Inspected, cleaned, organized, and prepared outdoor equipment to ensure safety, quality, and readiness.

Summer Camp Counselor | UC Davis Youth Programs **May-Aug 2026**
Davis, CA

- Supervised youth participants throughout daily camp activities, maintaining a safe, inclusive, and engaging environment.
- Led games, creative activities, and group transitions while adapting instruction to different ages, needs, and energy levels.
- Communicated with campers, coworkers, and families and responded calmly to behavioral, logistical, and safety concerns.

LEADERSHIP & COMMUNICATIONS

Social Media Director | UC Davis Women's Rowing **Apr 2023-Aug 2024**
Davis, CA

- Managed content across Instagram, Facebook, and TikTok, creating graphics, short-form video, newsletters, and event recaps to strengthen team visibility.
- Built collaborations with rowing programs and local organizations and used platform analytics to refine content strategy and engagement.

CURRENT GRADUATE COURSEWORK

CS-GY 6033 Design and Analysis of Algorithms I **Fall 2026 - In Progress**
Graduate study of algorithm design, correctness, complexity analysis, induction, asymptotics, heaps, and core problem-solving paradigms.

CS-GY 6373 Programming Languages **Fall 2026 - In Progress**
Graduate study of programming-language principles, paradigms, syntax and semantics, parsing, and implementation concepts.

CS-GY 6513 Big Data **Fall 2026 - In Progress**
Graduate study of systems and methods for processing large-scale datasets, including distributed-data workflows and frequent-pattern mining.

CURRENT ACADEMIC WORK

Semantic Joins and Value Matching Operators | Big Data Team Project **Fall 2026 - In Progress**
Team: Game of Joins

- Working in a three-person team to investigate semantic joins and value-matching operators for identifying related values that may not match exactly.
- Developing a staged project plan covering problem definition, system approach, implementation, evaluation, and final presentation milestones.

ADDITIONAL UNDERGRADUATE COURSEWORK

Introduction to Programming (Python); Data Structures in Python; Data Structures in C; Artificial Intelligence; Computer Organization and Machine-Dependent Programming; Algorithm Design and Analysis; Theory of Computation; Discrete Mathematics; Calculus I-III; Differential Equations; Vector Calculus; Linear Algebra; MATLAB Laboratory; Probability and Statistical Modeling; Cognitive Neuroscience.

AREAS OF INTEREST

Machine learning and artificial intelligence; ML research; human-computer interaction; human-centered computing; data science; applied software and technical product development.